

Boundary Integral Equations

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1 Laplace Equation

1.1 Introduction

Much of the work in this section follows Folland [1], especially chapter 3. That book uses a different sign convention on the Laplace layer potentials, so some of the results look slightly different here.

First let's consider the Laplace equation on the interior/exterior with Dirichlet/Neumann data. Here Ω is a bounded domain with finitely many connected components $\Omega_1, \dots, \Omega_m$ and $\Gamma = \partial\Omega$. Denote the exterior region as $\Omega' = \mathbb{R}^n - \overline{\Omega}$, which has connected components $\Omega'_0, \Omega'_1, \dots, \Omega'_{m'}$ (Ω'_0 is the unbounded component).

$$\begin{aligned} \text{Interior Dirichlet: } \Delta u &= 0 \quad \text{in } \Omega \\ u &= f \quad \text{in } \Gamma \end{aligned}$$

$$\begin{aligned} \text{Interior Neumann: } \Delta u &= 0 \quad \text{in } \Omega \\ \frac{\partial u}{\partial n} &= g \quad \text{in } \Gamma \end{aligned}$$

$$\begin{aligned} \text{Exterior Dirichlet: } \Delta u &= 0 \quad \text{in } \Omega' \\ u &= f \quad \text{in } \Gamma \end{aligned}$$

$$\begin{aligned} \text{Exterior Neumann: } \Delta u &= 0 \quad \text{in } \Omega' \\ \frac{\partial u}{\partial n} &= g \quad \text{in } \Gamma \end{aligned}$$

$\Phi(x)$ is the free space fundamental solution to the Laplace equation.

$$\begin{aligned} \mathbb{R}^2 : \quad \Phi(x) &= -\frac{1}{2\pi} \log |x| \\ \mathbb{R}^3 : \quad \Phi(x) &= \frac{1}{4\pi|x|} \end{aligned}$$

We will also frequently use the normal derivative of $\Phi(x - y)$:

$$\begin{aligned} \mathbb{R}^2 : \quad \frac{\partial \Phi(x - y)}{\partial n_y} &= \frac{(x - y) \cdot n_y}{2\pi|x - y|^2} \\ \mathbb{R}^3 : \quad \frac{\partial \Phi(x - y)}{\partial n_y} &= \frac{(x - y) \cdot n_y}{4\pi|x - y|^3} \end{aligned}$$

Note that x and y are vectors in $\mathbb{R}^n$, but for convenience, I am not using vector notation. We additionally will work with 4 integral operators for the Laplace Equation:

$$\begin{aligned} \text{Single Layer: } S_\Gamma[\sigma](x) &:= \int_\Gamma \Phi(x - y) \sigma(y) dS_y \\ \text{Double Layer: } D_\Gamma[\sigma](x) &:= \int_\Gamma \frac{\partial \Phi(x - y)}{\partial n_y} \sigma(y) dS_y \\ \text{Adjoint Double Layer: } S'_\Gamma[\sigma](x) &:= \int_\Gamma \frac{\partial \Phi(x - y)}{\partial n_x} \sigma(y) dS_y \\ \text{Adjoint Single Layer: } D'_\Gamma[\sigma](x) &:= \int_\Gamma \frac{\partial^2 \Phi(x - y)}{\partial n_x \partial n_y} \sigma(y) dS_y \end{aligned}$$

1.2 The Interior Dirichlet Problem

Assume the solution can be written as a **double layer potential** for some unknown density $\sigma \in C(\Gamma)$.

$$u(x) = D_\Gamma[\sigma](x) = \int_\Gamma \frac{\partial \Phi(x-y)}{\partial n_y} \sigma(y) dS_y$$

Theorem. $u(x)$ is harmonic for $x \in \Omega$.

Proof. If $x \in \Omega$,

$$\begin{aligned} \Delta_x u(x) &= \Delta_x \int_\Gamma \frac{\partial \Phi(x-y)}{\partial n_y} \sigma(y) dS_y \\ &= \int_\Gamma \frac{\partial}{\partial n_y} \underbrace{(\Delta_x \Phi(x-y))}_{=0} \sigma(y) dS_y \\ &= 0 \quad \blacksquare \end{aligned}$$

It has also been shown that the double layer formulation obeys the following jump condition: For $x \in \Omega$ and $x_0 \in \Gamma$,

$$\lim_{x \rightarrow x_0} u(x) = -\frac{1}{2} \sigma(x_0) + \int_\Gamma \frac{\partial \Phi(x_0-y)}{\partial n_y} \sigma(y) dS_y$$

Appendix A contains a derivation of this jump condition in 2 dimensions, using complex analysis. Since we want to solve the interior Dirichlet problem, we will enforce the boundary condition by setting

$$f(x) = -\frac{1}{2} \sigma(x) + \int_\Gamma \frac{\partial \Phi(x-y)}{\partial n_y} \sigma(y) dS_y, \quad x \in \Gamma \quad (1)$$

Thus, in order to solve the problem, we have to solve equation 1 for the unknown density σ . We could equivalently write this equation, which is a Fredholm equation of the second kind, more compactly as

$$\left(-\frac{1}{2} I + D_\Gamma \right) [\sigma] = f. \quad (2)$$

1.3 The Exterior Dirichlet Problem

We again assume the solution can be written as a **double layer potential**.

$$u(x) = D_\Gamma[\sigma](x) = \int_\Gamma \frac{\partial \Phi(x-y)}{\partial n_y} \sigma(y) dS_y$$

The proof that u is harmonic in Ω' holds again since $\Phi(x-y)$ is harmonic for all $x \neq y$. Now we wish to enforce the boundary condition from the exterior. We have a similar jump condition as before: For $x \in \Omega'$ and $x_0 \in \Gamma$,

$$\lim_{x \rightarrow x_0} u(x) = \frac{1}{2} \sigma(x_0) + \int_\Gamma \frac{\partial \Phi(x_0-y)}{\partial n_y} \sigma(y) dS_y$$

So we then enforce the boundary condition by setting

$$f(x) = \frac{1}{2} \sigma(x) + \int_\Gamma \frac{\partial \Phi(x-y)}{\partial n_y} \sigma(y) dS_y, \quad x \in \Gamma \quad (3)$$

Which we can also write as

$$\left(\frac{1}{2} I + D_\Gamma \right) [\sigma] = f. \quad (4)$$

1.4 The Interior Neumann Problem

Assume the solution can be written as a **single layer potential** for some unknown density $\sigma \in C(\gamma)$.

$$u(x) = \int_{\Gamma} \Phi(x-y)\sigma(y)dS_y$$

Theorem. $u(x)$ is harmonic for $x \in \Omega$.

Proof. If $x \in \Omega$,

$$\begin{aligned} \Delta_x u(x) &= \Delta_x \int_{\Gamma} \Phi(x-y)\sigma(y)dS_y \\ &= \int_{\Gamma} \Delta_x \Phi(x-y)\sigma(y)dS_y \\ &= 0 \quad \blacksquare \end{aligned}$$

In order to enforce the Neumann boundary condition $\frac{\partial u}{\partial n} = g$ on Γ , we need to know how the normal derivative of the single layer formulation behaves as we approach the boundary from the interior of Ω . Note that the normal derivative of $u(x) = S_{\Gamma}[\sigma](x)$ is another one of the integral formulations introduced at the beginning: the adjoint double layer $\frac{\partial u}{\partial n_x} = S'_{\Gamma}[\sigma](x)$. It turns out that we get a similar jump condition for the adjoint double layer. For $x \in \Omega$, $x_0 \in \Gamma$, we have

$$\lim_{x \rightarrow x_0} \frac{\partial u(x)}{\partial n_x} = \frac{1}{2}\sigma(x_0) + \int_{\Gamma} \frac{\partial \Phi(x_0-y)}{\partial n_x} \sigma(y)dS_y$$

Thus, enforcing the Neumann boundary condition gives us

$$g(x) = \frac{1}{2}\sigma(x) + \int_{\Gamma} \frac{\partial \Phi(x-y)}{\partial n_x} \sigma(y)dS_y, \quad x \in \Gamma \quad (5)$$

Or equivalently,

$$\left(\frac{1}{2}I + S'_{\Gamma} \right) [\sigma] = g. \quad (6)$$

1.5 The Exterior Neumann Problem

We again assume the solution can be written as a **single layer potential**.

$$u(x) = \int_{\Gamma} \Phi(x-y)\sigma(y)dS_y$$

The proof that u is harmonic on Ω' holds for the same reason as in the interior problem, since if $x \in \Omega'$ and $y \in \Gamma$, then $x \neq y$. We get a similar jump condition for the adjoint double layer as before. For $x \in \Omega'$, $x_0 \in \Gamma$, we have

$$\lim_{x \rightarrow x_0} \frac{\partial u(x)}{\partial n_x} = -\frac{1}{2}\sigma(x_0) + \int_{\Gamma} \frac{\partial \Phi(x_0-y)}{\partial n_x} \sigma(y)dS_y$$

Thus, enforcing the Neumann boundary condition gives us

$$g(x) = -\frac{1}{2}\sigma(x) + \int_{\Gamma} \frac{\partial \Phi(x-y)}{\partial n_x} \sigma(y)dS_y, \quad x \in \Gamma \quad (7)$$

Or equivalently,

$$\left(-\frac{1}{2}I + S'_{\Gamma} \right) [\sigma] = g. \quad (8)$$

1.6 Solvability

We want to solve the four integral equations

$$\begin{aligned} -\frac{1}{2}\sigma + D_\Gamma[\sigma] &= f & (\text{Int. Dirichlet}) \\ \frac{1}{2}\sigma + D_\Gamma[\sigma] &= f & (\text{Ext. Dirichlet}) \\ \frac{1}{2}\sigma + S'_\Gamma[\sigma] &= g & (\text{Int. Neumann}) \\ -\frac{1}{2}\sigma + S'_\Gamma[\sigma] &= g & (\text{Ext. Neumann}) \end{aligned}$$

where $\sigma \in C(\Gamma)$. Without going deep into analysis, note that this will generally extend to $\sigma \in L^2(\Gamma)$, which can be approximated arbitrarily well by a continuous function.

D_Γ and S'_Γ are compact operators, and Fredholm theory is applicable. We are then concerned with the null space of each left hand side:

$$\begin{aligned} V_+ &= \left\{ \sigma \mid D_\Gamma[\sigma] = \frac{1}{2}\sigma \right\} & (\text{Int. Dirichlet}) \\ V_- &= \left\{ \sigma \mid D_\Gamma[\sigma] = -\frac{1}{2}\sigma \right\} & (\text{Ext. Dirichlet}) \\ W_- &= \left\{ \sigma \mid S'_\Gamma[\sigma] = -\frac{1}{2}\sigma \right\} & (\text{Int. Neumann}) \\ W_+ &= \left\{ \sigma \mid S'_\Gamma[\sigma] = \frac{1}{2}\sigma \right\} & (\text{Ext. Neumann}) \end{aligned}$$

There is an important result that shows up in Appendix A, but is more general. We have

$$\begin{aligned} \int_{\partial\Omega} \frac{\partial\Phi(x-y)}{\partial n_y} dS_y &= \begin{cases} 0, & x \in \Omega' \\ -\frac{1}{2}, & x \in \Gamma \\ -1, & x \in \Omega \end{cases} \\ \int_{\partial\Omega'} \frac{\partial\Phi(x-y)}{\partial n_y} dS_y &= \begin{cases} 0, & x \in \Omega' \\ \frac{1}{2}, & x \in \Gamma \\ 1, & x \in \Omega \end{cases} \end{aligned}$$

Thus, if we define the functions

$$\begin{aligned} \alpha_j(x) &= \begin{cases} 1, & x \in \partial\Omega_j \\ 0, & \text{otherwise} \end{cases}, \quad j = 1, \dots, m \\ \alpha'_j(x) &= \begin{cases} 1, & x \in \partial\Omega'_j \\ 0, & \text{otherwise} \end{cases}, \quad j = 1, \dots, m' \end{aligned}$$

we see that $\alpha_j \in V_+$ and $\alpha'_j \in V_-$ for each $j = 1, \dots, m$. The set of α_j functions is linearly independent, as well as the set of α'_j functions ($\langle \alpha_j, \alpha_k \rangle = 0$ if $j \neq k$. Same for α'_j functions). Thus $\dim V_+ \geq m$ and $\dim V_- \geq m'$. Also, since D_Γ and S'_Γ are adjoints of one another, we have $\dim V_+ = \dim W_+$ and

$$\dim V_- = \dim W_-.$$

(Leaving out the details for now) It turns out that $\dim V_+ = \dim W_+ = m$, and

- $n > 2$: for each $(a_1, \dots, a_m) \in \mathbb{C}^m$, there is a unique $\beta \in W_+$ such that $w = S_\Gamma[\beta]$ satisfies $w|_{\Omega_j} = a_j$ for $j = 1, \dots, m$.
- $n = 2$: there is an $(m - 1)$ dimensional subspace X of $\mathbb{C}^m$ such that
 - $\mathbb{C}^m = X \oplus \mathbb{C}(1, \dots, 1)$
 - for each $(a_1, \dots, a_m) \in X$ there is a unique $\beta \in W_+^0 = \{\beta \mid \int_\Gamma \beta = 0\}$ such that $w = S_\Gamma[\beta]$ satisfies $w|_{\Omega_j} = a_j$ for $j = 1, \dots, m$.

We have slightly different result for $n = 2$ that arises from the fact that $\Phi(z)$ is unbounded in $\mathbb{R}^2$.

The result is simpler for V_- and W_- , because if $\beta \in W_-$, it necessarily satisfies $\int_\Gamma \beta = 0$. Thus we have $\dim V_- = \dim W_- = m'$ and for each $(a_1, \dots, a_{m'}) \in \mathbb{C}^{m'}$, there is a unique $\beta \in W_+$ such that $w = S_\Gamma[\beta]$ satisfies $w|_{\Omega_j} = a_j$ for $j = 1, \dots, m'$.

1.7 Main Results

It turns out that it is sufficient to use

$$\int_\Gamma \frac{1}{|x|^{n-2}} \sigma(y) dS_y$$

in place of the single layer to complete the formulation for interior and exterior Dirichlet problems [2]. This is of course preferable because the integrand is no longer singular on Γ , and can be easily computed. Then, if $f, g \in C(\Gamma)$, the main results are as follows.

1.7.1 Interior Dirichlet Solution

- The solution exists and is unique.
- The solution in $\mathbb{R}^n$ is representable as

$$u = D_\Gamma[\sigma] + \int_\Gamma \frac{1}{|x|^{n-2}} \sigma(y) dS_y$$

where $D_\Gamma[\sigma]$ solves the interior Dirichlet problem

- Enforcing the boundary conditions gives us the Fredholm of the 2nd kind equation

$$-\frac{1}{2}\sigma + D_\Gamma[\sigma] + \int_\Gamma \frac{1}{|x|^{n-2}} \sigma(y) dS_y = f.$$

1.7.2 Exterior Dirichlet Solution

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- Enforcing the boundary conditions gives us the Fredholm of the 2nd kind equation

$$\frac{1}{2}\sigma + D_\Gamma[\sigma] + \int_\Gamma \frac{1}{|x|^{n-2}} \sigma(y) dS_y = f.$$

1.7.3 Interior Neumann Solution

- The solution exists if and only if $\int_{\partial\Omega_j} f = 0$ for $j = 1, \dots, m$.
- The solution is unique up to functions constant on each Ω_j , for $j = 1, \dots, m$.
- The solution is representable as

$$u = S_\Gamma[\sigma]$$

- Enforcing the boundary conditions gives us the Fredholm of the 2nd kind equation

$$\frac{1}{2}\sigma + S'_\Gamma[\sigma] = g$$

1.7.4 Exterior Neumann Solution

- The solution exists if and only if $\int_{\partial\Omega'_j} f = 0$ for $j = 1, \dots, m'$. If $n = 2$, we additionally need this to hold for the unbounded component $\int_{\partial\Omega'_0} f = 0$.
- The solution is unique up to functions constant on each Ω'_j , for $j = 1, \dots, m'$. If $n = 2$, then Ω'_0 is also included.
- The solution is representable as

$$u = S_\Gamma[\sigma]$$

- Enforcing the boundary conditions gives us the Fredholm of the 2nd kind equation

$$-\frac{1}{2}\sigma + S'_\Gamma[\sigma] = g$$

2 The Stokes Equation

This section largely follows the work in Pozrikidis [3]. The governing equations for an incompressible Newtonian fluid are

$$\begin{aligned} -\nabla P + \mu \nabla^2 \mathbf{u} + \rho \mathbf{b} &= \mathbf{0} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

where $\mathbf{u}$ is the fluid velocity, P is the pressure, ρ is the constant density, μ is the viscosity, and $\mathbf{b}$ is the constant body force. The first equation arises from Newton's 2nd Law, and the second arises from mass conservation (the continuity equation).

we may also instead write the body force term in terms of a stress tensor σ :

$$\rho \mathbf{b} = \nabla P - \mu \nabla^2 \mathbf{u} = \nabla \cdot \boldsymbol{\sigma}$$

where

$$\sigma_{ij} \equiv -P\delta_{ij} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right). \quad (9)$$

2.1 The Reciprocal Identity

We have a useful identity that is used frequently, known as the reciprocal identity. Let $\mathbf{u}$ and $\mathbf{u}'$ be two solutions to the Stokes flow equation, with associated stress tensors $\boldsymbol{\sigma}$ and $\boldsymbol{\sigma}'$. Then we have

$$\nabla \cdot (\mathbf{u}' \cdot \boldsymbol{\sigma} - \mathbf{u} \cdot \boldsymbol{\sigma}') \quad (10)$$

In index notation, this is

$$\frac{\partial}{\partial x_j} (u'_i \sigma_{ij} - u_i \sigma'_{ij})$$

and if we integrate over a volume V with boundary D and apply the divergence theorem, we get

$$\int_D \mathbf{u}' \cdot \mathbf{f} dS = \int_D \mathbf{f}' \cdot \mathbf{u} dS$$

where $\mathbf{f} = \boldsymbol{\sigma} \cdot \mathbf{n}$.

2.2 The Stokeslet and Stresslet

The free space fundamental solution to the Stokes flow equation is $\mathbf{u}$ that solves

$$\begin{aligned} -\nabla P + \mu \nabla^2 \mathbf{u} + \mathbf{g} \delta(\mathbf{x} - \mathbf{y}) &= \mathbf{0} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

where $\mathbf{g}$ is an arbitrary constant vector, $\mathbf{y}$ is an arbitrary point, and δ is the 3D delta function. The solution to this problem is

$$u_i(\mathbf{x}) = G_{ij}(\mathbf{x}, \mathbf{y}) g_j$$

where $\mathbf{G}$ is the free-space Green's function, or Stokeslet:

$$G_{ij}(\mathbf{x}, \mathbf{y}) = \frac{1}{8\pi\mu} \left(\frac{\delta_{ij}}{|\mathbf{x} - \mathbf{y}|} + \frac{(x_i - y_i)(x_j - y_j)}{|\mathbf{x} - \mathbf{y}|^3} \right). \quad (11)$$

This solution is interpreted physically as the flow of a concentrated point force of strength $\mathbf{g}$ located at $\mathbf{y}$. The pressure and stress fields associated with the flow $\mathbf{u}$ can be written as

$$\begin{aligned} P(\mathbf{x}) &= p_j(\mathbf{x}, \mathbf{y}) g_j \\ \sigma_{ik}(\mathbf{x}) &= T_{ijk}(\mathbf{x}, \mathbf{y}) g_j \end{aligned}$$

where $\mathbf{p}$ is the pressure vector, and $\mathbf{T}$ is the Stresslet. The explicit formula for the pressure vector is

$$p_i(\mathbf{x} - \mathbf{y}) = \frac{x_i - y_i}{4\pi|\mathbf{x} - \mathbf{y}|^3}.$$

Thus, using equation 9 we obtain the Stresslet

$$T_{ijk}(\mathbf{x}, \mathbf{y}) = -\frac{3}{4\pi} \left(\frac{(x_i - y_i)(x_j - y_j)(x_k - y_k)}{|\mathbf{x} - \mathbf{y}|^5} \right) \quad (12)$$

2.3 Boundary Integral Formulation

To transition into the boundary integral formulation, we use the reciprocal identity 10 with $u'_i(\mathbf{x}) = G_{ij}(\mathbf{x}, \mathbf{y}) g_j$, which implies $\sigma'_{ik}(\mathbf{x}) = T_{ijk}(\mathbf{x}, \mathbf{y}) g_j$. Here $\mathbf{x}$ is a "source point" in the region V . Inserting these, and then integrating over V and applying divergence theorem, we have

$$\int_D [G_{ij}(\mathbf{x}, \mathbf{y}) \sigma_{ik}(\mathbf{y}) - \mu u_i(\mathbf{y}) T_{ijk}(\mathbf{x}, \mathbf{y})] n_k(\mathbf{y}) dS(\mathbf{y}) = 0 \quad (13)$$

Equivalently:

$$\int_D G_{ij}(\mathbf{x}, \mathbf{y}) f_i(\mathbf{y}) dS(\mathbf{y}) - \mu \int_D u_i(\mathbf{y}) T_{ijk}(\mathbf{x}, \mathbf{y}) n_k(\mathbf{y}) dS(\mathbf{y}) = 0$$

Using equation 13, but now with the volume $V - B_\epsilon(\mathbf{x})$ (ϵ is small enough so that it is disjoint from D), we have

$$\begin{aligned} & \int_D [G_{ij}(\mathbf{x}, \mathbf{y}) \sigma_{ik}(\mathbf{y}) - \mu u_i(\mathbf{y}) T_{ijk}(\mathbf{x}, \mathbf{y})] n_k(\mathbf{y}) dS(\mathbf{y}) \\ &= - \int_{\partial B_\epsilon(\mathbf{x})} [G_{ij}(\mathbf{x}, \mathbf{y}) \sigma_{ik}(\mathbf{y}) - \mu u_i(\mathbf{y}) T_{ijk}(\mathbf{x}, \mathbf{y})] n_k(\mathbf{y}) dS(\mathbf{y}) \end{aligned}$$

Leaving out the analysis required to prove this result, it turns out that as $\epsilon \rightarrow 0$, the Green's function and stress tensor are equal to the Stokeslet and Stresslet to leading order in ϵ , and the integral reduces simply to $u_j(\mathbf{x})$. We therefore have the boundary integral representation of $\mathbf{u}$:

$$u_j(\mathbf{x}) = - \int_D G_{ij}(\mathbf{x}, \mathbf{y}) f_i(\mathbf{y}) dS(\mathbf{y}) + \mu \int_D u_i(\mathbf{y}) T_{ijk}(\mathbf{x}, \mathbf{y}) n_k(\mathbf{y}) dS(\mathbf{y}) \quad (14)$$

The first integral is the **single layer**, and the second integral is the **double layer**. In general, for some boundary densities ϕ and ψ , the single and double layer integral operators are defined as

$$\begin{aligned} \mathcal{S}[\phi](\mathbf{x})_i &= \int_D G_{ij}(\mathbf{x}, \mathbf{y}) \phi_j(\mathbf{y}) dS(\mathbf{y}) \\ \mathcal{D}[\psi](\mathbf{x})_i &= \int_D T_{ijk}(\mathbf{x}, \mathbf{y}) n_k(\mathbf{y}) \psi_j(\mathbf{y}) dS(\mathbf{y}) \end{aligned}$$

We will also make use of the **traction kernel**, which is simply the normal derivative of the single layer:

$$\mathcal{K}[\phi](\mathbf{x})_i = \int_D T_{ijk}(\mathbf{x}, \mathbf{y}) n_k(\mathbf{x}) \phi_j(\mathbf{y}) dS(\mathbf{y})$$

2.4 Jump Conditions

In Dirichlet and Neumann problems, our boundary conditions will be prescribed surface velocities or prescribed surface forces at the boundary. We obtain similar jump conditions as in the Laplace equation boundary integral formulations. For convenience, the $\pm$ superscript is used to indicate the limit from the exterior/interior in the normal direction, and the $\int^{\mathcal{PV}}$ symbol indicates a principal value integral; that is, the integral over the boundary minus a ball of radius ϵ centered at the singularity, in the limit as $\epsilon \rightarrow 0$.

- Single Layer

- The velocity $u = \mathcal{S}[\phi]$ is continuous across the boundary D .
- The surface force $\frac{\partial u}{\partial n_x} = \mathcal{K}[\phi]$ obeys the following jump conditions

$$\begin{aligned} f_i^+(\mathbf{x}) &\equiv \sigma_{ik}^+(\mathbf{x}) n_k(\mathbf{x}) = -\frac{1}{2} \phi_i(\mathbf{x}) + \int_D^{\mathcal{PV}} T_{ijk}(\mathbf{x}, \mathbf{y}) n_k(\mathbf{x}) \phi_j(\mathbf{y}) dS(\mathbf{y}) \\ f_i^-(\mathbf{x}) &\equiv \sigma_{ik}^-(\mathbf{x}) n_k(\mathbf{x}) = \frac{1}{2} \phi_i(\mathbf{x}) + \int_D^{\mathcal{PV}} T_{ijk}(\mathbf{x}, \mathbf{y}) n_k(\mathbf{x}) \phi_j(\mathbf{y}) dS(\mathbf{y}) \end{aligned}$$

- Double Layer

- The velocity $u = \mathcal{D}[\psi]$ obeys the following jump conditions

$$\begin{aligned} u_i^+(\mathbf{x}) &= \frac{1}{2} \psi_i(\mathbf{x}) + \int_D^{\mathcal{PV}} T_{ijk}(\mathbf{x}, \mathbf{y}) n_k(\mathbf{y}) \psi_j(\mathbf{y}) dS(\mathbf{y}) \\ u_i^-(\mathbf{x}) &= -\frac{1}{2} \psi_i(\mathbf{x}) + \int_D^{\mathcal{PV}} T_{ijk}(\mathbf{x}, \mathbf{y}) n_k(\mathbf{y}) \psi_j(\mathbf{y}) dS(\mathbf{y}) \end{aligned}$$

(b) The surface force is continuous across the boundary D .

2.5 The Interior Dirichlet Problem

2.6 The Exterior Dirichlet Problem

2.7 The Interior Neumann Problem

The solution to the interior Neumann problem exists only if the force and torque on D from $\mathbf{f}^-$ are zero. Conveniently, this constraint corresponds to rigid body motion, which is our application of interest. Then, for rigid body motion, the solution density is

$$\psi = \psi^p + \sum_{i=1}^6 c_i \psi'^{(i)}.$$

ψ^p is a particular solution, c_i are arbitrary coefficients, and $\psi'^{(i)}$ correspond to the surface forces external to D for the six modes of rigid body motion.

2.8 The Exterior Neumann Problem

The solution to the exterior Neumann problem exists as long as

$$\int_D u_i(\mathbf{y}) n_i(\mathbf{y}) ds(\mathbf{y}) = 0$$

Meaning the flow rate through the boundary is zero. Then the general solution is

$$\psi = \psi^p + c\mathbf{n}$$

where c is an arbitrary coefficient and $\mathbf{n}$ is the normal vector.

A a proof of the D_Γ jump condition in 2D.

Coming soon.

References

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